

Thiago Ferreira dos Santos

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Last update: August 2026

RESEARCH INTERESTS

Stellar Astrophysics. Pop. III, stellar abundances, oscillations, activity, white dwarfs, wide and close binary systems;

Exoplanets and Brown Dwarfs; evolution and star-planet interactions.

Galactic Structure and Archaeology. stellar surveys, Galactic bulge and disc, stellar content, and infrared observations.

EDUCATION

PhD candidate in Astrophysics – Yale University, US 2023 – Present

MSc in Astronomy – Universidade de São Paulo, Brazil 2021 – 2023

BSc in Physics – Universidade Federal de Santa Catarina (UFSC), Brazil 2016 – 2021

REFEREED PUBLICATIONS

Aug. 2026: **15** peer-reviewed (**7** as 1st-author) || h-index: **7** | Citations: **136** | ADS – [Full List](#) and [Refereed](#) | [Google Scholar](#)

15. **Ferreira, T.** & Rice, M., [AJ](#), **172**, 111 (2026)

On the Eccentricity Distribution and Tidal Evolution of Transiting Brown Dwarfs

14. **Ferreira, T.**, Bellinger, E.P., Farag, E., Lindsay, C.J., [ApJ](#), **1001**, 41 (2026)

Red-Giant Asteroseismology of Low-Mass Population III Stars

13. **Ferreira, T.**, Bellinger, E.P., Farag, E., Lindsay, C.J., [ApJ](#), **1000**, 147 (2026)

Evolution of Low-Mass Population III Stars: Convection, Mass Loss, Nucleosynthesis, and Neutrinos

12. **Ferreira, T.**, Yana Galarza, J., et al., [accepted for publication in AJ](#) (2026); [arXiv:2512.17072](#)

Hidden Companions of the Early Milky Way I. New α -Enhanced Exoplanet Hosts

11. Winnick, I., et al. (incl. **Ferreira, T.**), [accepted for publication in AJ](#) (2025); [arXiv:2508.16513](#)

Extreme Lithium Depletion in Solar Twins: Challenging Non-Standard Mixing Models

10. Galarza, J.Y., Lorenzo-Oliveira, D., **Ferreira, T.**, et al., [ApJ](#), **983**, 70 (2025)

HIP 8522: A Puzzling Young Solar Twin with the Lowest Detected Lithium

9. Fermiano, V., et al. (incl. **Ferreira, T.**), [A&A](#), **690**, L7 (2024)

The Young Exoplanetary System TOI-4562: Confirming The Presence of a Third Body in the System

8. **Ferreira, T.**, Rice, M., et al., [AJ](#), **168**, 145 (2024)

SOLES XI. The Aligned Orbit of TOI-2533 b, a Transiting Brown Dwarf Orbiting an F8-type Star

7. Galarza, J.Y., **Ferreira, T.**, Lorenzo-Oliveira, D., et al., [AJ](#), **168**, 91 (2024)

TOI-1173 A b: The First Inflated Super-Neptune in a Wide Binary System

6. Galarza, J.Y., Reggiani, H., **Ferreira, T.**, et al., [ApJ](#), **974**, 122 (2024)

Detailed Abundances of the Planet-Hosting TOI-1173 A/B: Possible Evidence of Planet Engulfment in a Very Wide Binary

5. Saito, R.K., et al. (incl. **Ferreira, T.**), [A&A](#), **689**, A148 (2024)

The VISTA Variables in the Vía Láctea eXtended (VVVX) ESO Public Survey: Completion of the Observations and Legacy

4. **Ferreira, T.**, Saito, R.K., Minniti, D., et al., [MNRAS](#), **527**, 10737 (2024)

A Benchmark White Dwarf–Ultracool Dwarf Wide Field Binary

3. Botan, E., et al. (incl. **Ferreira, T.**), [MNRAS](#), **504**, 654 (2021)

Unveiling Short-Period Binaries in the Inner VVV Bulge

2. Herpich, F.R., et al. (incl. **Ferreira, T.**), [A&A, 647, A169 \(2021\)](#)

VVV Survey Near-Infrared Colour Catalogue of Known Variable Stars

1. **Ferreira, T.**, Saito, R.K., Minniti, D., et al., [MNRAS, 486, 1220 \(2019\)](#)

The Asymptotic Evolution of the Stellar Merger V1309 Sco: a Blue Straggler in the Making?

TELESCOPE OBSERVING EXPERIENCE

ESO VLT / ESPRESSO – 1 night, Co-I	2026B (FTC 117A)
Gemini North/South 8.1-m / GHOST & MAROON-X – 1 night, Co-I	2026B
Keck I 10-m / HIRES – 2 nights, PI	2026A
Subaru 8.2-m / HDS – 1.5 nights, PI	2026A
Gemini South 8.1-m / GHOST – 1.5 nights, PI	2026A
Magellan Clay 6.5-m / MIKE & PFS – 2 nights, Co-I	2026A
Gemini North/South 8.1-m / GHOST & MAROON-X – 2 nights, Co-I	2026A
Gemini South 8.1-m / GHOST – 0.5 nights, DDT	2025B
Magellan Clay 6.5-m / WINERED – 1 night, PI	2025A
Magellan Clay 6.5-m / PFS – 2 nights, Co-I	2025A
Magellan Clay 6.5-m / MIKE – 12 nights, Co-I	2024B, 2025A
ESO 3.6-m / HARPS – 6 nights, PI	2023A (P111)

SEMINARS

Asteroseismology of Low-Mass Population III Stars – TOS group, HITS/Heidelberg (Germany)	Jul 2025
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CONFERENCE TALKS & POSTERS

10th TESS/17th Kepler Asteroseismic Science Consortium Workshop, Aarhus, Denmark Red-Giant Asteroseismology of Low-Mass Population III Stars (talk). Robust Mass, Radius, and Age Measurement of a Nearly Pristine Star (poster; Grunblatt et al.).	Jul 2026
Cool Stars 23, Tokyo, Japan Robust Mass, Radius, and Age Measurement of a Nearly Pristine Star (poster; Grunblatt et al.). A Homogeneous Census of Thick Disc Planetary Systems (poster).	Jun 2026
9th TESS/16th Kepler Asteroseismic Science Consortium Workshop, Vienna, Austria On the Structure and Oscillations of the First Stars (poster).	Jul 2025
exoLaTam 22, Santiago, Chile HIP 102152b: A Low-Mass Planet Candidate Around an Old Solar Twin (talk).	Dec 2022
XLV Annual Meeting of the Brazilian Astronomical Society, Virtual HIP 102152b: A Low-Mass Planet Candidate Around an Old Solar Twin (talk).	Oct 2022
Cool Stars 21, Virtual HIP 102152b: A Low-Mass Planet Candidate Around an Old Solar Twin (poster).	Jul 2022
Exoplanets III, Virtual A Box-Fitting Implementation for the Transit Method Detection of Exoplanets (poster).	Jul 2021
Sagan Exoplanet Summer Workshop, Virtual Time-Series Analysis of Near-IR Variabilities on VVV M Dwarf Stars (poster).	Jul 2021
ALMA Planets2020 Workshop, Santiago Exoplanetary Transits in the Galactic Bulge (poster).	Mar 2020

- EWASS 2019, Lyon, France Jun 2019
Infrared Counterparts of Gaia Alerts (poster; Ivanov et al.).
- VVV & VVV-X Science Workshop, Florianópolis, Brazil Apr 2018
On the Photometric Study of the Cataclysmic Variable IGR J17014-4306 (poster).
- Seminário de Iniciação Científica, UFSC, Virtual 2018, 2019, 2020, 2021
Stellar Variability Towards the Central Region of the Milky Way with the VVV/VVV-X Surveys (talk).
- PPGFSC/UFSC Meeting, Florianópolis, Brazil 2018, 2020
Stellar Variability Towards the Central Region of the Milky Way with the VVV/VVV-X Surveys (poster).

FELLOWSHIPS, AWARDS & GRANTS

Yale Conference Travel Fund - Newton Endowment	Jul 2025, Jul 2026
GSAS Yale Graduate Fellowship	2023 – Present
Boris Garfinkel Prize Fellowship in Astronomy	2023
CAPES M.Sc. Research Fellowship, IAG/USP, Brazil	2021 – 2023
exoLaTam 2022 Travel Award	Dec 2022
Teaching Improvement Program, IAG/USP	2022
ESO/ALMA Planets2020 Workshop Travel Award	Mar 2020
CNPq Undergraduate Research Fellowship, UFSC, Brazil	2017 – 2021

WORKSHOPS & ADVANCED SCHOOLS

YCIU Primordial Black Holes Workshop, US	Mar 2026
Yale AstroCodEx Hack Day, New Haven	Aug 2025
MESA Summer School, Leuven, Belgium	Jul 2025
Yale Tinsley Workshop, New Haven	Oct 2024
New York Area Exoplanets Meeting, New York	May 2024
ESO Star–Planet Connection Workshop, Virtual	Oct 2021
XIII CBPF School, Virtual	Aug 2021
ZTF Summer School on Variable Stars, Virtual	Aug 2021
Sagan Exoplanet Summer Workshop, Virtual	Jul 2021

SERVICE AND TEACHING

Referee – ApJ	2025
Teaching Assistant, ASTR 110, 220, 310, 556 – Yale University	Aug 2024 – May 2026
Teaching Assistant, AGA 0502 – IAG/USP	2022

ACADEMIC REFERENCES

Earl P. Bellinger (PhD advisor) – Yale University	earl.bellinger@yale.edu
Jorge L. Meléndez (MSc advisor) – Universidade de São Paulo	jorge.melendez@iag.usp.br
Roberto K. Saito (BSc advisor) – Universidade Federal de Santa Catarina	robsaito@gmail.com
Dante Minniti – Universidad Andres Bello Vatican Observatory	vvvdante@gmail.com
